

Machine learning for data analytics

Assignment 6: Advanced Deep Learning

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Each assignment should be submitted as a Jupyter notebook with complete code, visualizations, and written analysis.

Problem 6.1: Deep Reinforcement Learning (40 points)

Implement DQN for game playing:

- Train agent on CartPole-v1 and LunarLander-v2 environments.
- Implement experience replay and target network.
- Track episode rewards, loss, and epsilon decay.
- Compare with PPO algorithm using stable-baselines3

Problem 6.2: Transfer Learning (30 points)

Fine-tune pre-trained models:

- Use ResNet50 pre-trained on ImageNet.
- Fine-tune on a small custom image dataset (e.g., Cats vs. Dogs with 500 samples per class).
- Experiment with freezing different numbers of layers.
- Compare with training from scratch.

Problem 6.3: Few-Shot Learning (30 points)

Implement prototypical networks:

- Use Omniglot or mini-ImageNet dataset.
- Implement 5-way 1-shot and 5-way 5-shot classification.
- Calculate episode-wise accuracy.
- Visualize embeddings using t-SNE.